

CSharPN

A Code-First Framework for Coloured Petri Net Modelling
with C# as Inscription Language

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To a younger generation of developers, Coloured Petri Nets look increasingly exotic.

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Underneath it, the older gap between **academia and industry**.

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CSharPN — a framework for working with Coloured Petri Nets in C#, with a VS Code extension for visualising and executing models.

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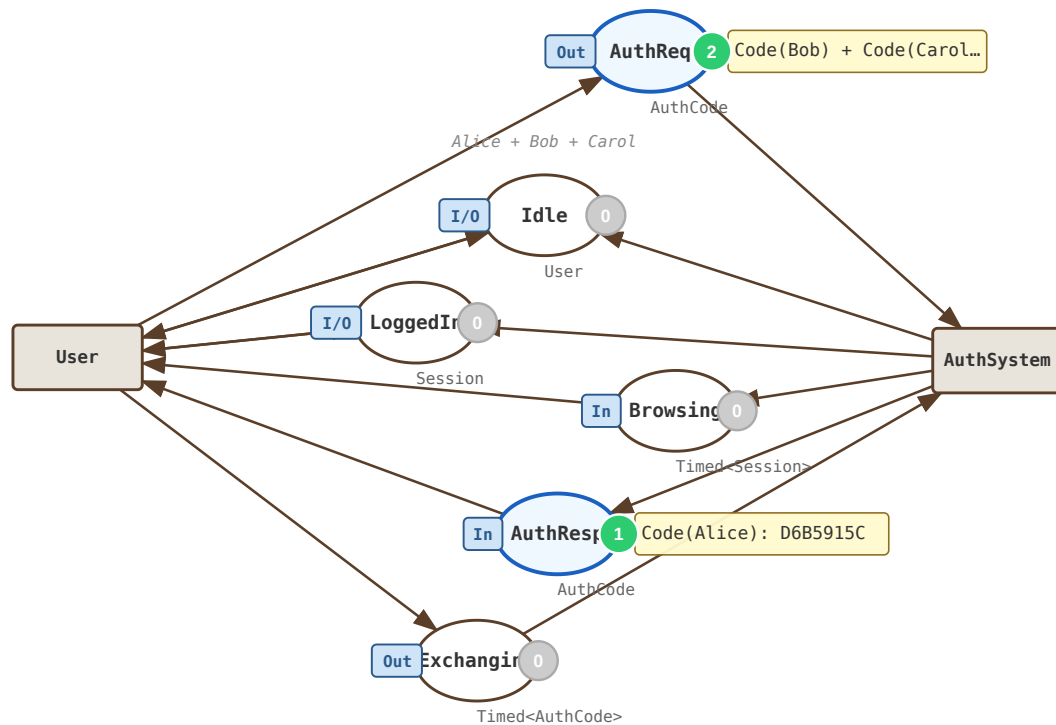
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Simulation-based development — with concurrency, resource sharing and timing modelled *explicitly*.

Running example: OAuth2 authoriz

Exercises all three features:

Top page: shared places link the two substitution transitions.

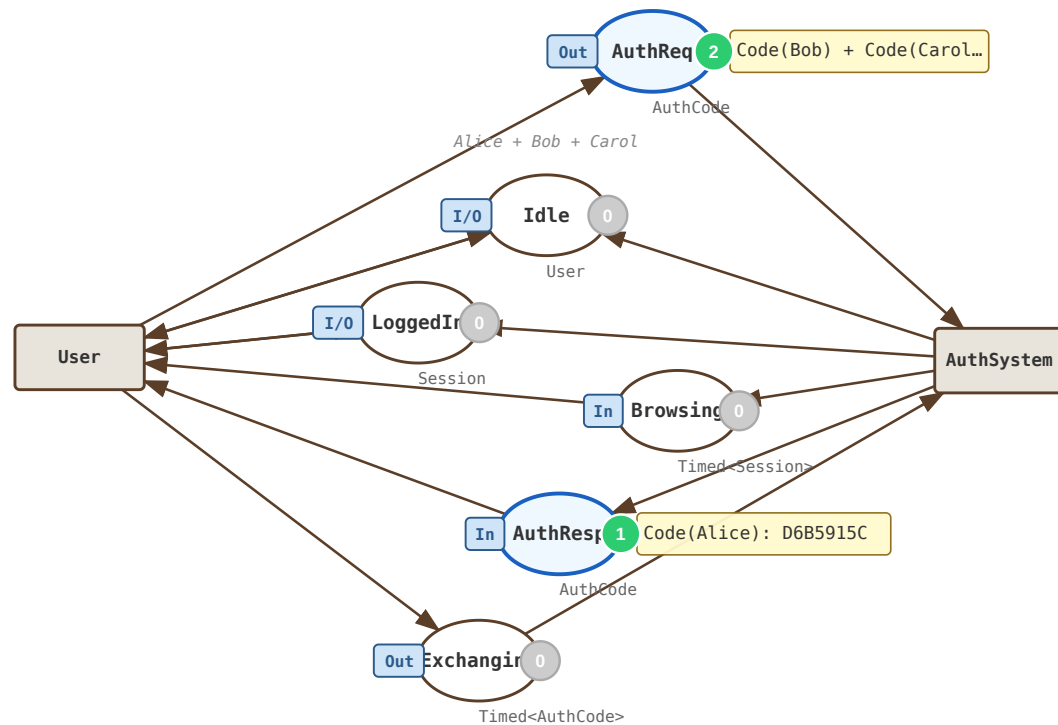


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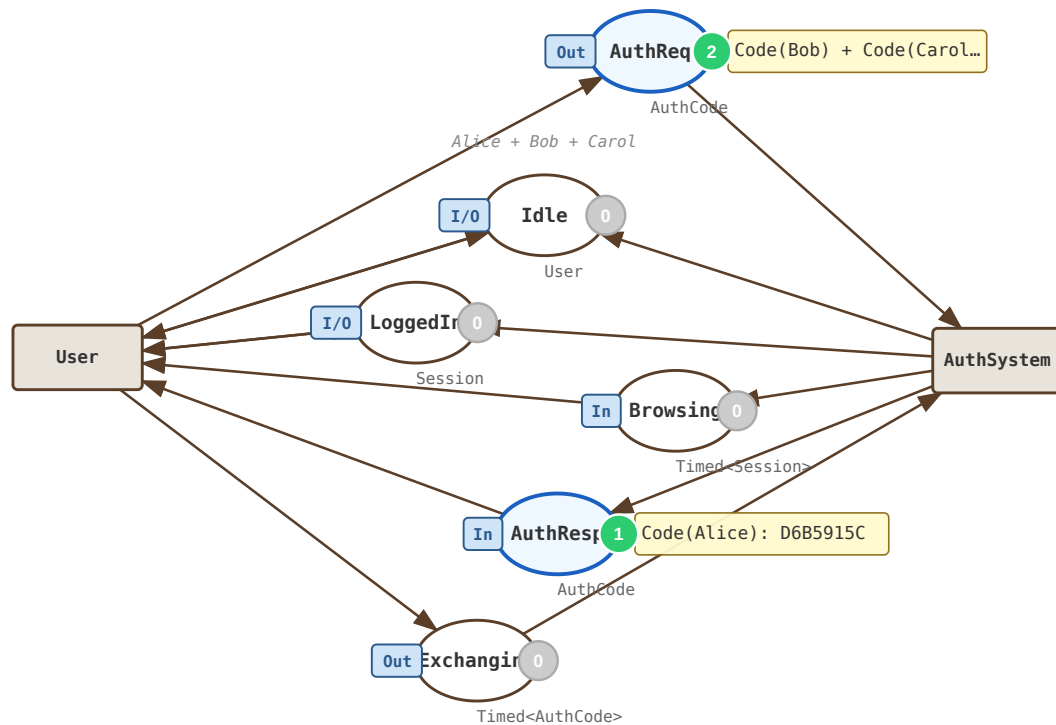


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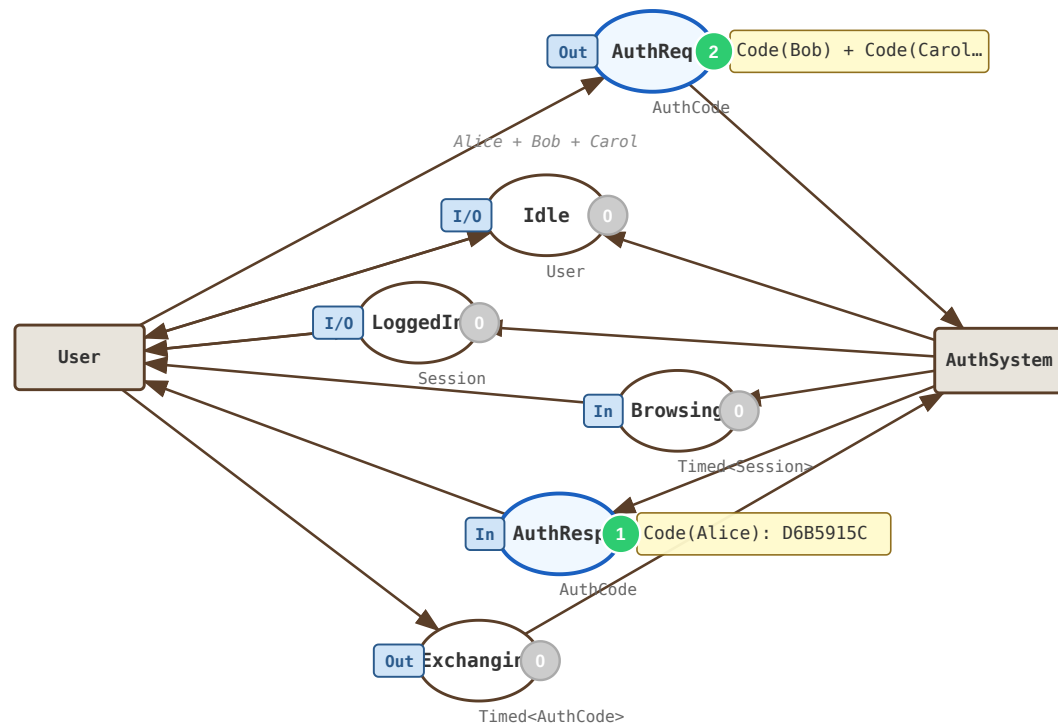


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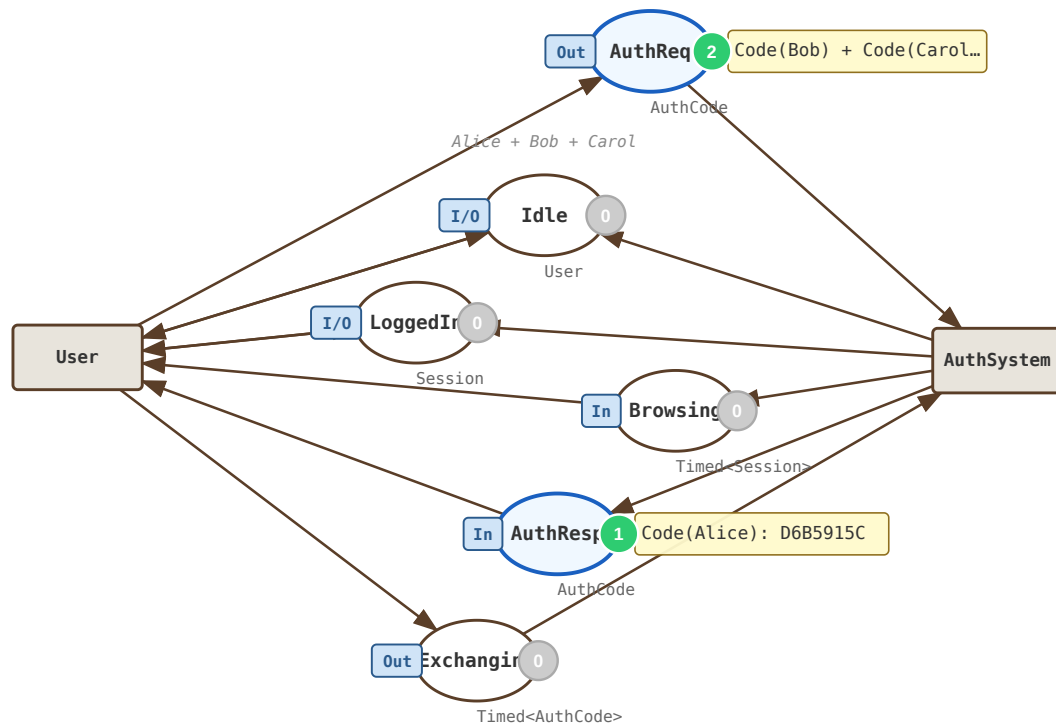


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- Timed auth, exchange, expiry
- Two sub-pages: User & AuthSystem

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Colour sets are just C# types

Any `notnull, IEquatable<T>` type works — primitives, enums, records, classes.

```
public record User(int Id, string Name) {  
    public string CsrfsState ⇒ $"s{Id}";  
    public int    BrowseTime ⇒ Id * 5;  
}  
  
public record AuthCode(User User, string Code, string State) {  
    public bool ValidCsrfs(User u) ⇒ State == u.CsrfsState;  
    public static AuthCode Create(User u, int seq) {  
        var hash = SHA256.HashData(Encoding.UTF8.GetBytes($"{u.Id}:{seq}"));  
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Places & multisets

```
// inside the model constructor
var authReq = AddPlace<AuthCode>("AuthReq");           // empty initial marking

var idle    = AddPlace("Idle", Multiset.Of(
    new User(1, "Alice"), new User(2, "Bob"), new User(3, "Carol")));

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Transitions: a fluent builder

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var u = new Var<User>("u");

AddTransition("Login")
    .Input(idle, u).Output(waiting, u)
    .Output(authReq, () => AuthCode.Create(u.Val), "AuthReq")
    .Build();

AddTransition("ReceiveCode")
    .Input(authResp, c).Input(waiting, u)
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Timed CPNs

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AddTransition("GrantToken")
    .TimedInput(exchg, c).Input(nextTokId, tid)
    .Output(nextTokId, () => tid.Val + 1)
    .Output(loggedIn, () => new Session(c.Val.User, tid.Val))
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    .Build();

AddTransition("SessionExpire")
    .TimedInput(expiry, se).Input(loggedIn, se2)
    .Guard(() => se.Val.TokenId == se2.Val.TokenId)
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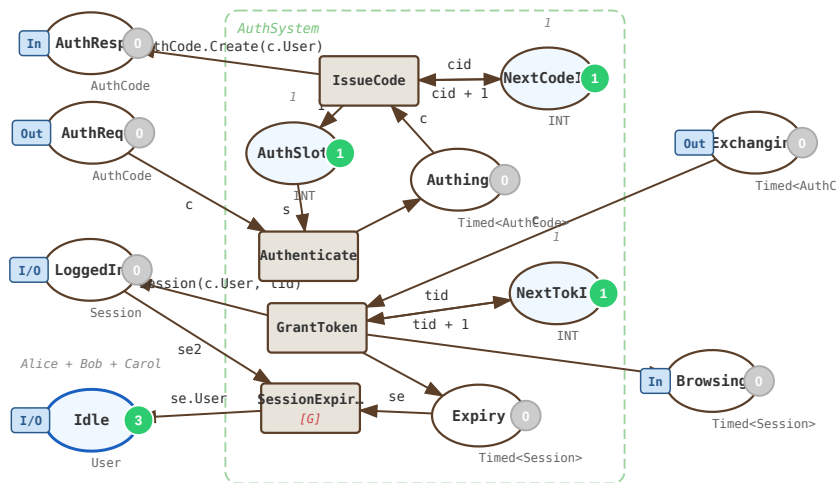
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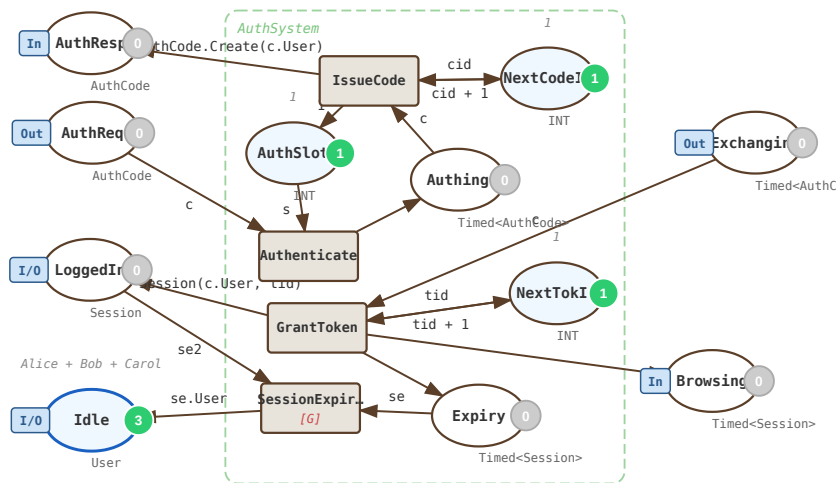
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class UserPage : CpnPage {
    public UserPage(Place<User> idle,
        Place<AuthCode> authReq, ...)
        : base("User") {
        InOut(idle);
        Out(authReq); In(authResp);
        Out(exchg);
        InOut(loggedIn); In(browsing);
        var waiting = AddPlace<User>("Waiting");
        // Login, ReceiveCode, Logout
    }
}
// assembly
AddSubPage(new UserPage(...), "User");
AddSubPage(new AuthSystemPage(...), "AuthSystem");
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AuthSystem sub-page: AuthSlot limits concurrency, Expiry delays timeout.

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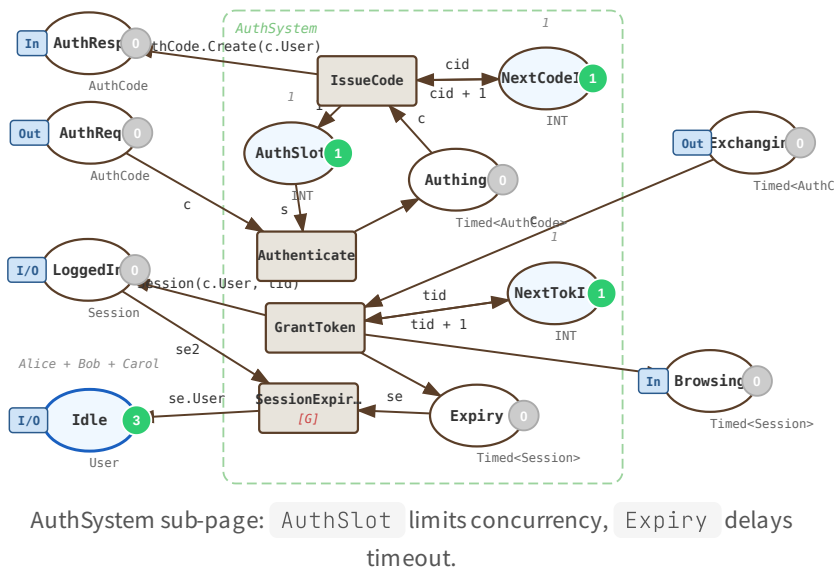


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- A sub-page is a CpnPage subclass

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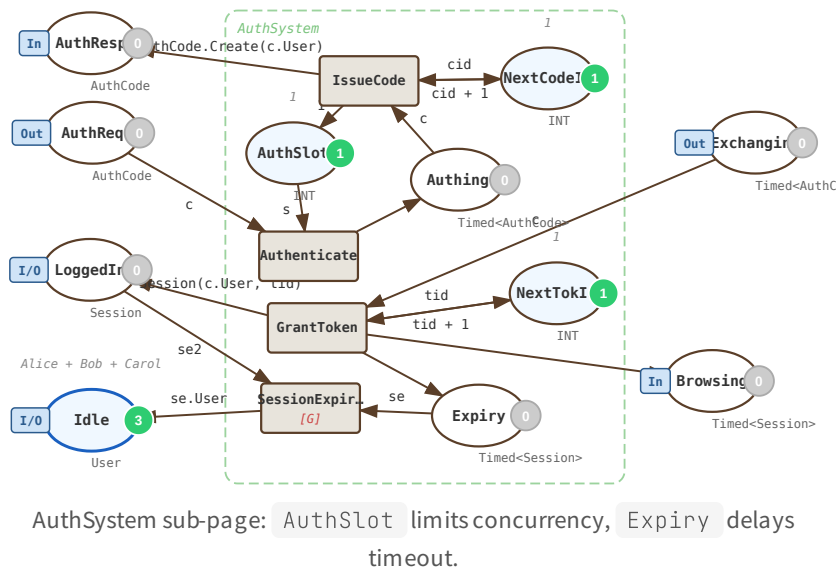
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        : base("User") {
        InOut(idle);
        Out(authReq); In(authResp);
        Out(exchg);
        InOut(loggedIn); In(browsing);
        var waiting = AddPlace<User>("Waiting");
        // Login, ReceiveCode, Logout
    }
}
// assembly
AddSubPage(new UserPage(...), "User");
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```



- A sub-page is a `CpnPage` **subclass**
- Ports use `In()` / `Out()` / `InOut()` — CPN Tools style

Hierarchical CPNs

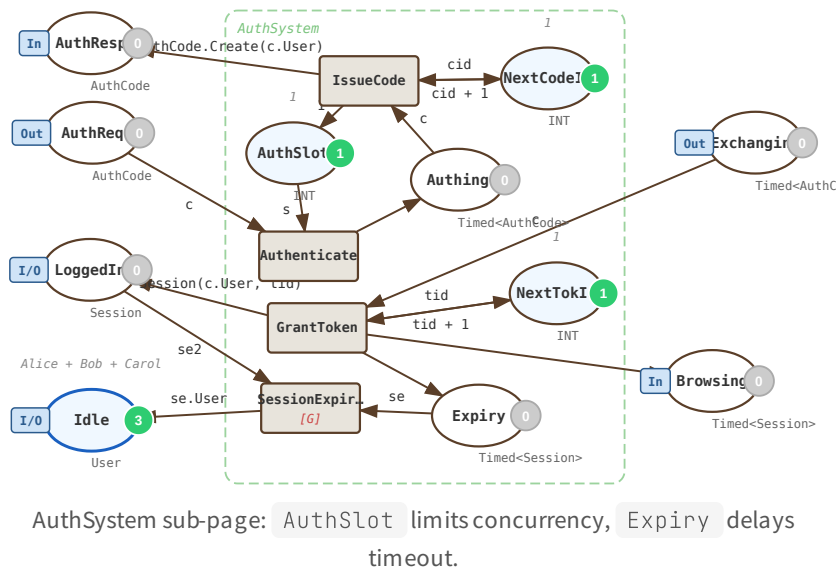
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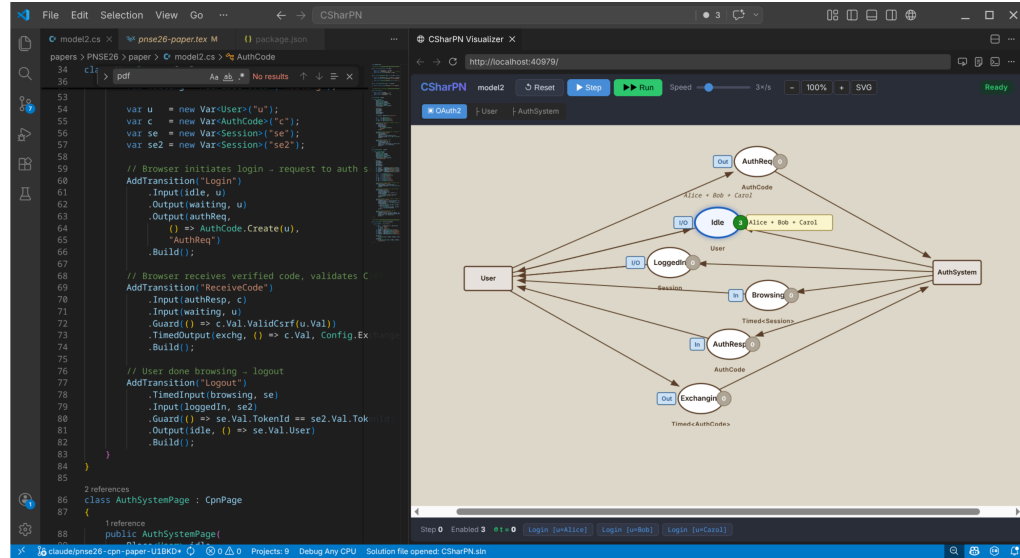
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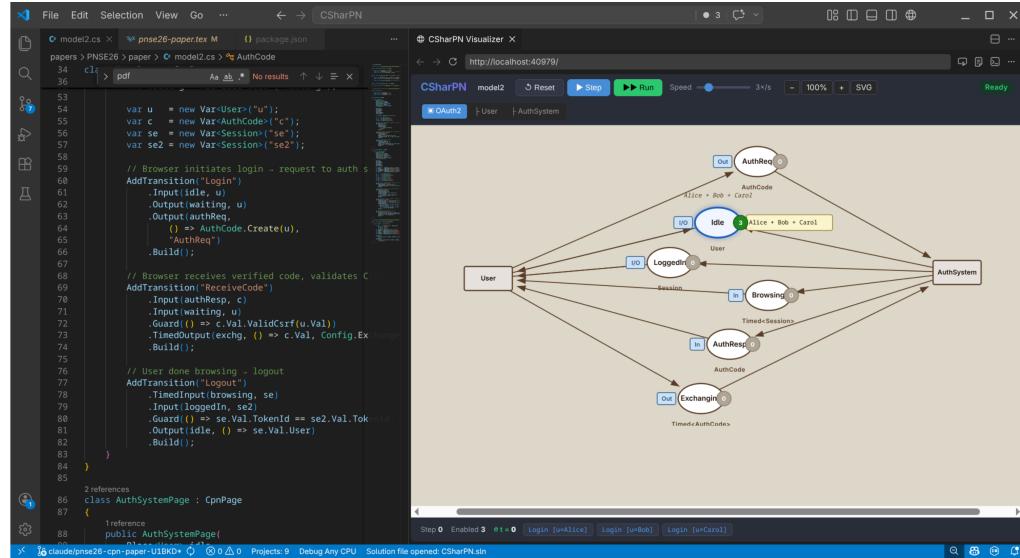
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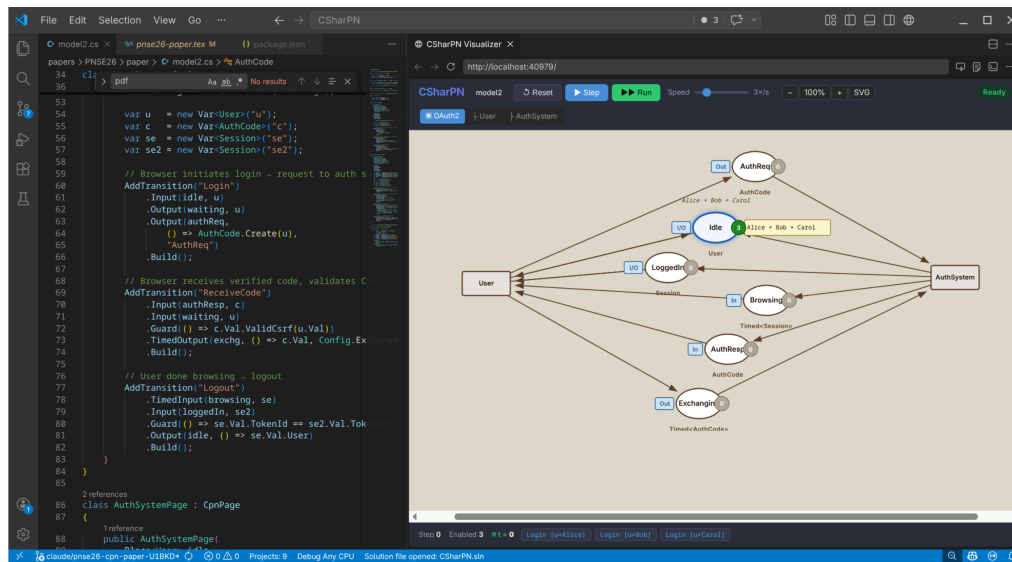
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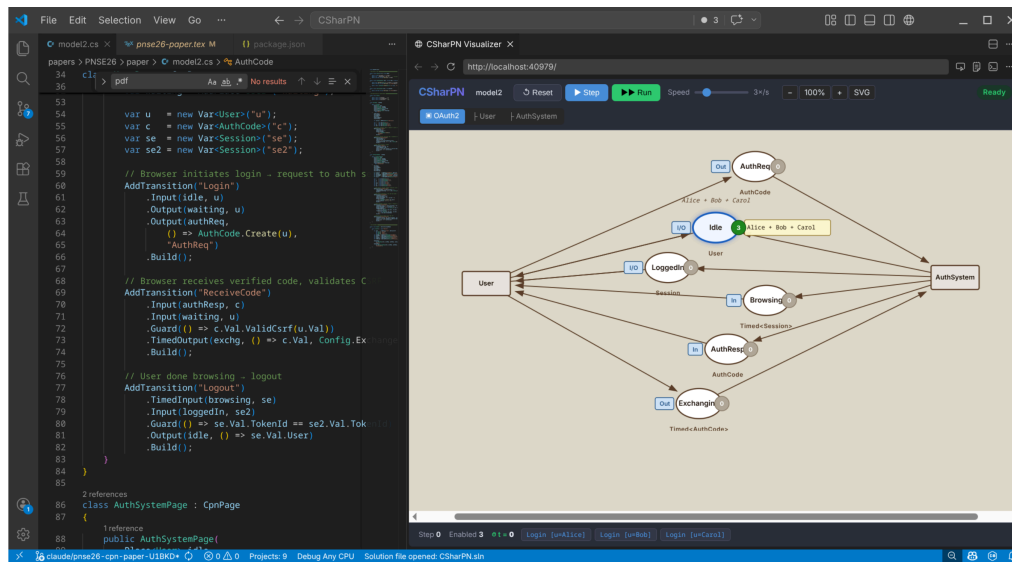
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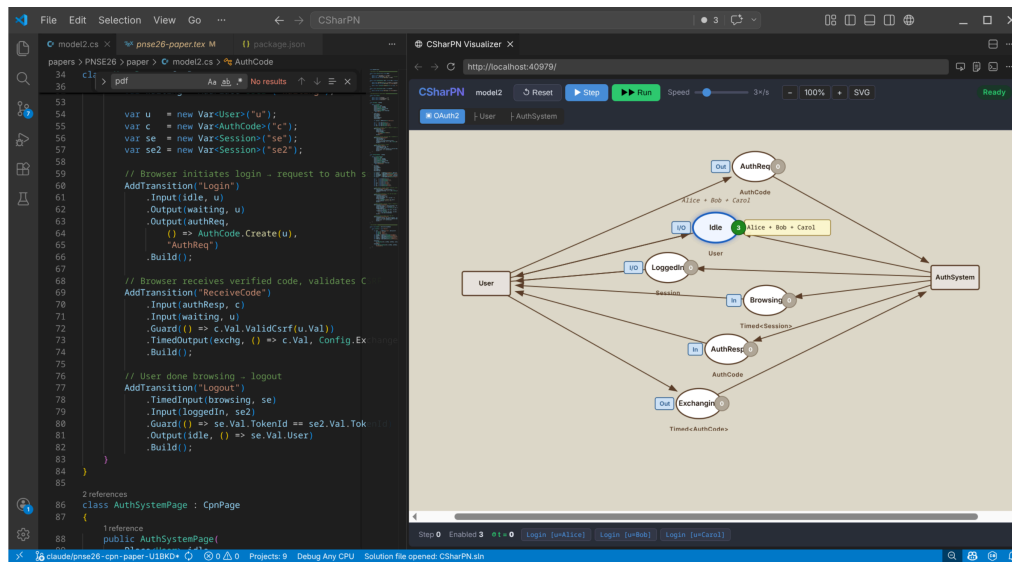
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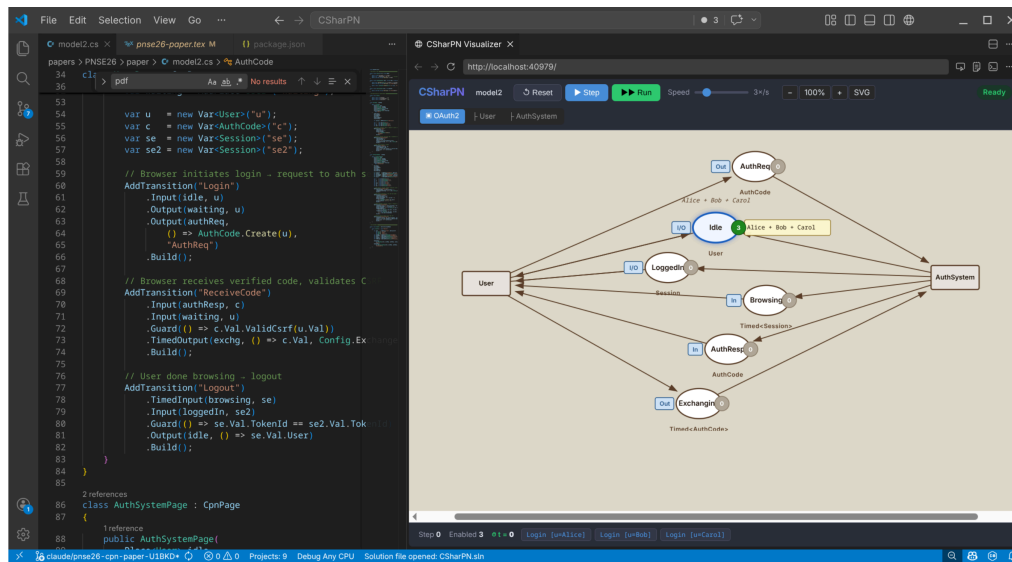
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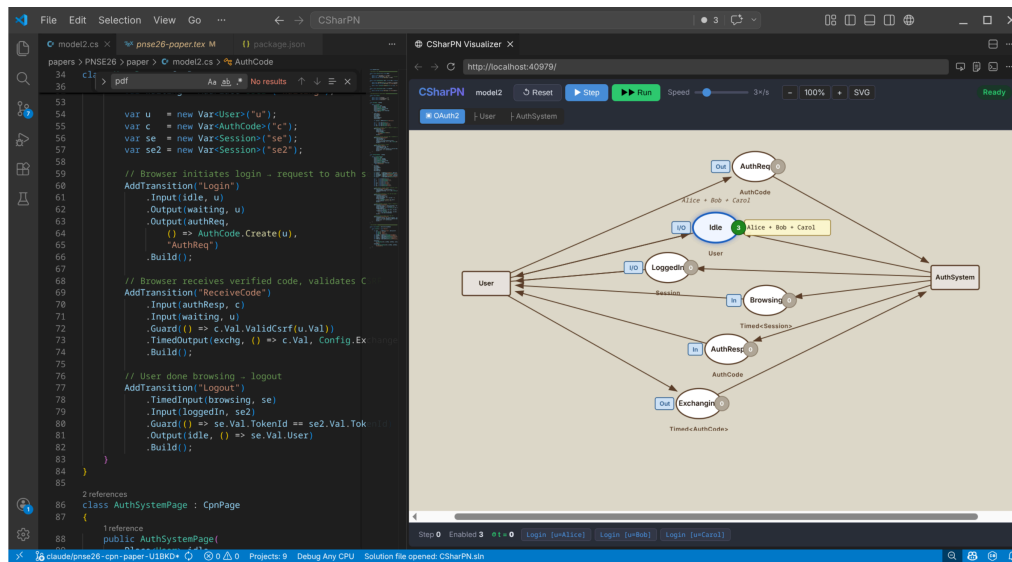
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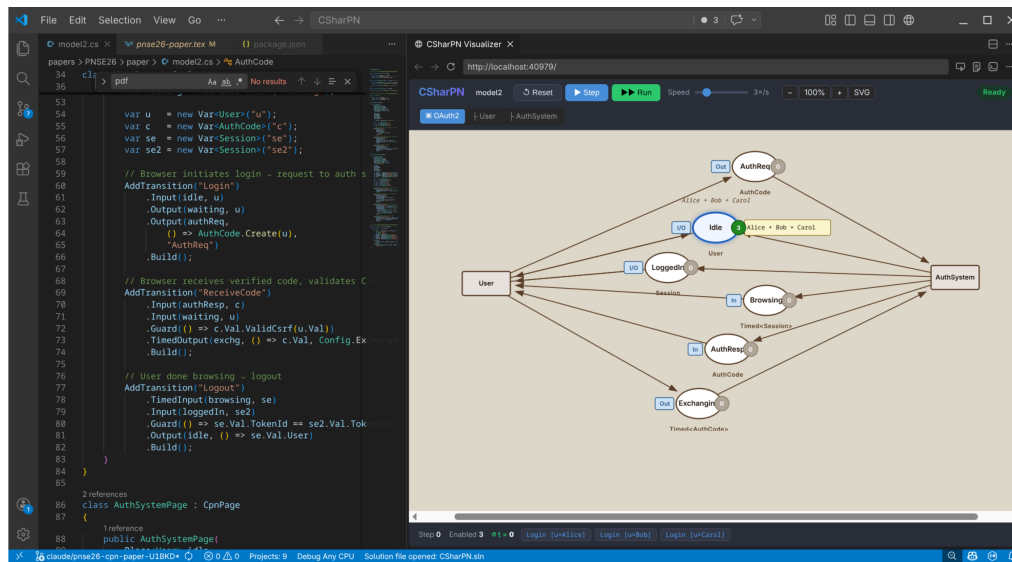
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- **Hot reload**: save .cs → recompile → reload, marking preserved



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key insight

When the inscription language is the host language, the model *is* the program.

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No code-generation step. No generated artefact, no model/implementation sync problem — the gap is zero by construction. (Caveat:

only when the deployment target is .NET.)

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Takeaway: when the inscription language is C#, the full power of a mature programming language — type system, libraries, tooling — is available to the CPN modeller at no extra cost.

Thank you

Questions?



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`github.com/simontjell/CSharpN.public`

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